

Q13

Vocabulary Richness Analysis

*"Which Discussion and Question Type
Produces the Richest Student Vocabulary?"*

5 Discussions · 2 Questions · 294 Responses

Data Understanding
& Preprocessing

Pattern Mining

Outlier Detection

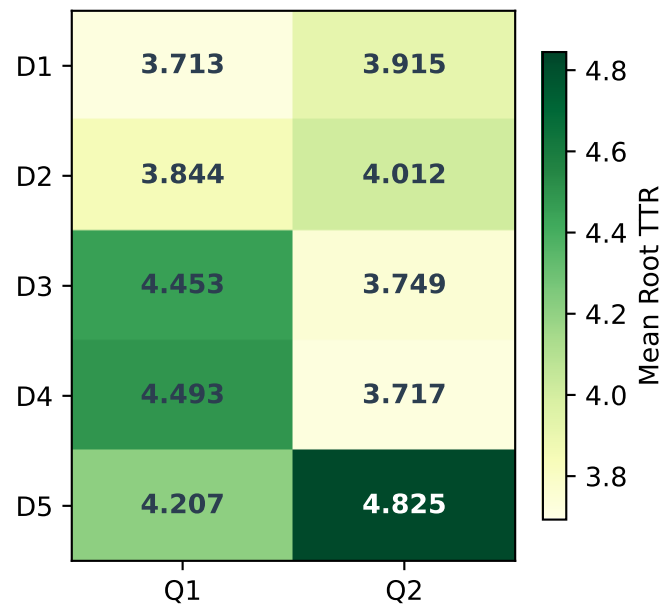
Metric: Root TTR = $\text{unique words} / \sqrt{(\text{total words})}$

More length-stable than plain TTR · Higher = richer vocabulary · Lower = more repetitive

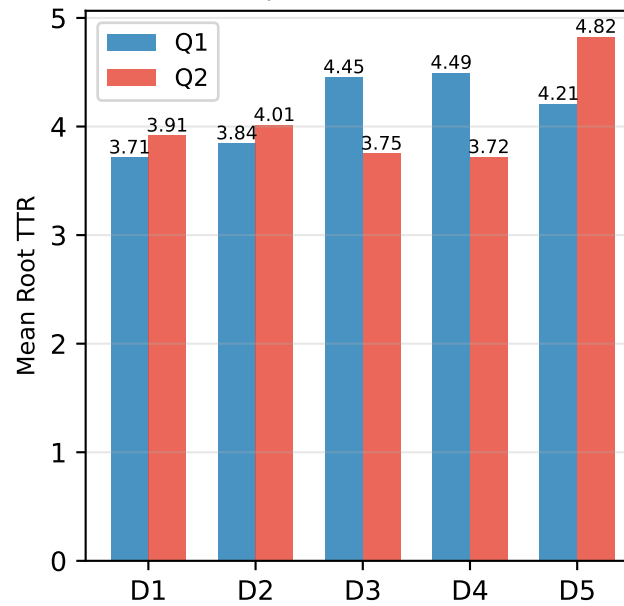
Preprocessing: "NOISY" labels removed · Min 2 tokens required

Data Understanding: Vocabulary Richness (Root TTR) Across Discussions & Questions

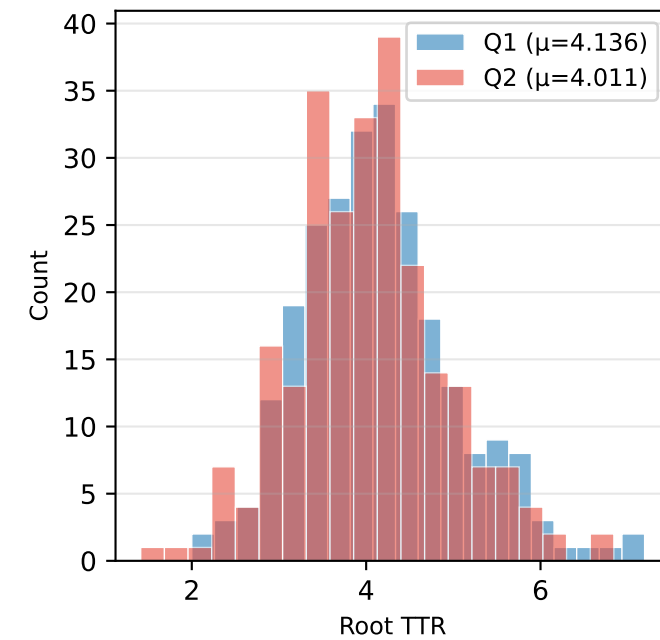
(A) Mean Root TTR Heatmap
(5 Discussions × 2 Questions)



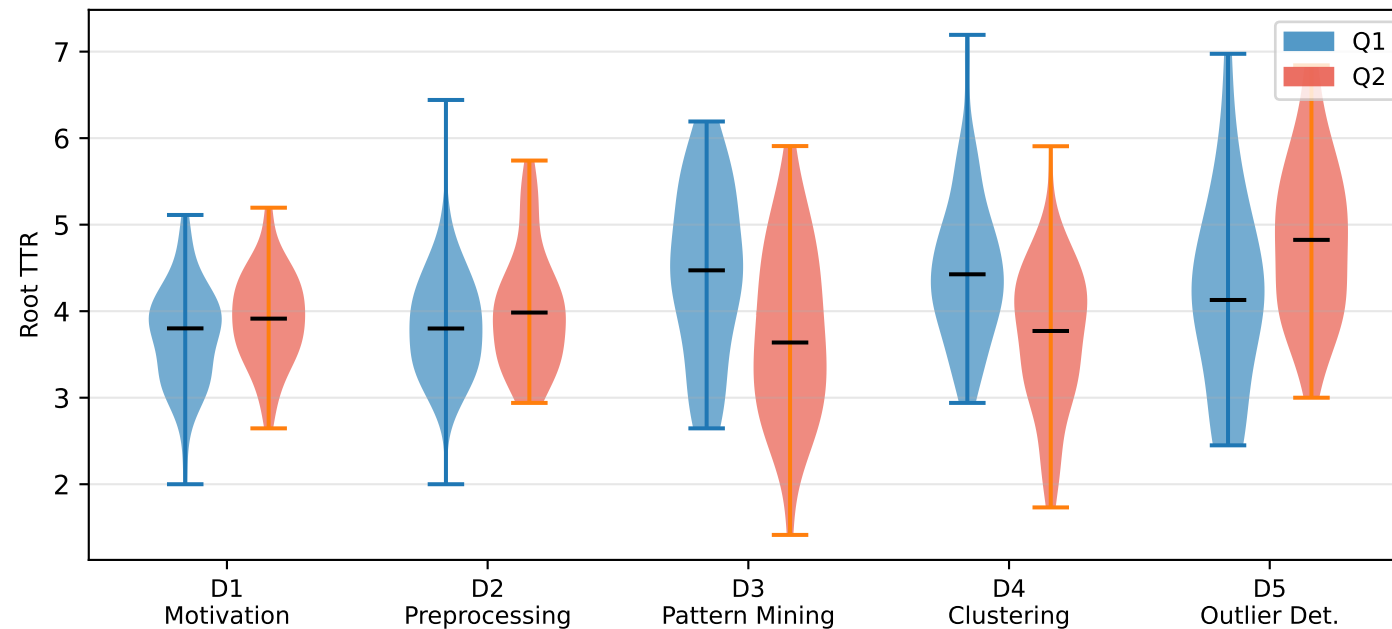
(B) Mean Root TTR: Q1 vs Q2
per Discussion



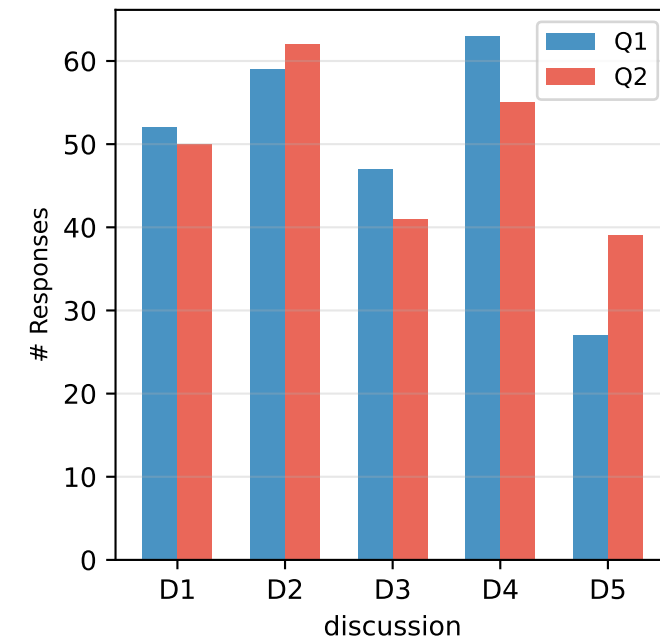
(C) Root TTR Distribution
Q1 vs Q2 (All Discussions)



(D) Root TTR Distribution per Discussion (Blue=Q1, Red=Q2)



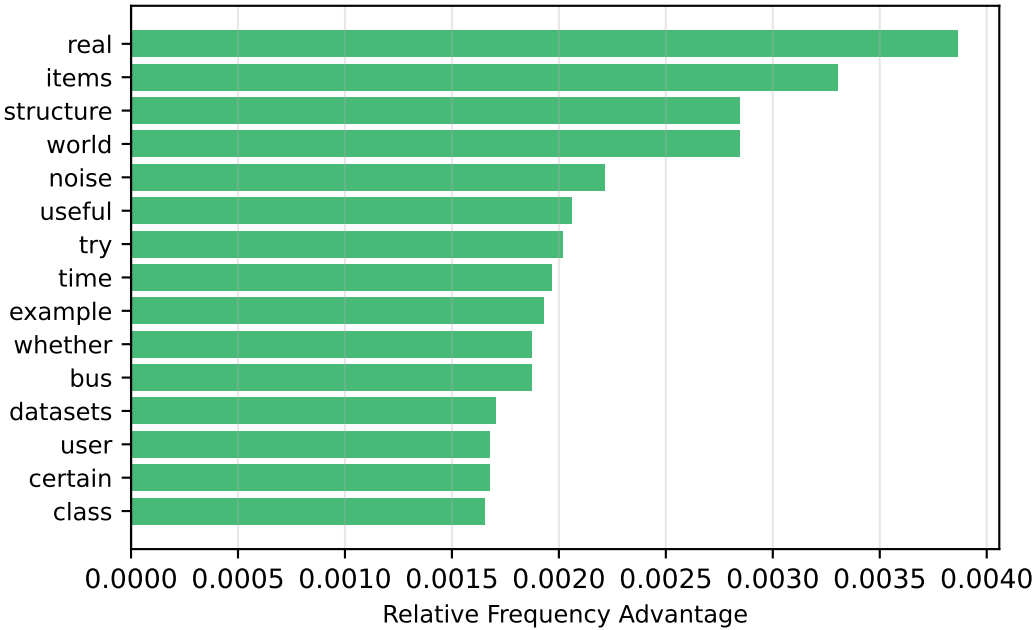
(E) Response Count
per Discussion × Question



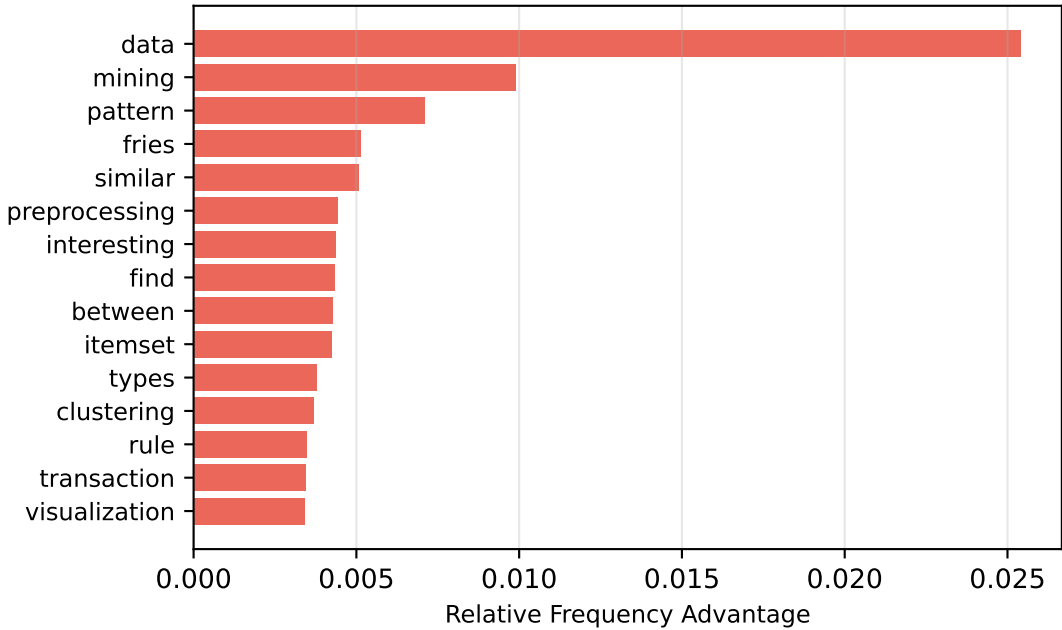
Key Finding: D5-Q2 has the highest mean Root TTR (4.825). Q2 responses tend to be richer than Q1 — evaluation/comparison questions require broader vocabulary.

Pattern Mining: What Linguistic Patterns Distinguish Rich vs. Poor Vocabulary Responses?

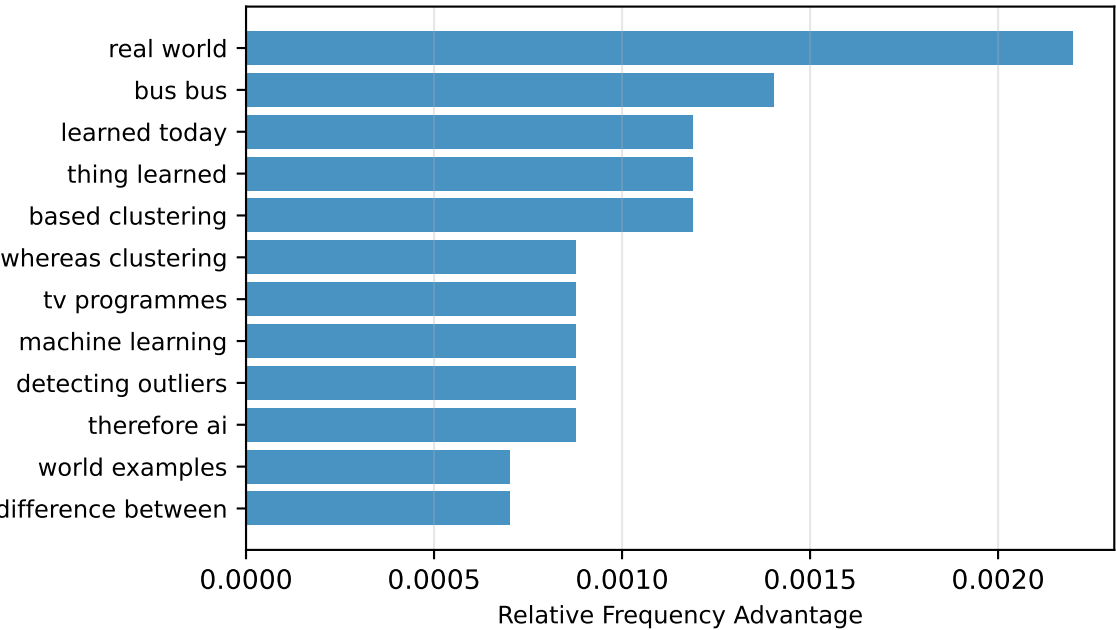
(A) Words More Common in HIGH Root TTR
(Rich Responses)



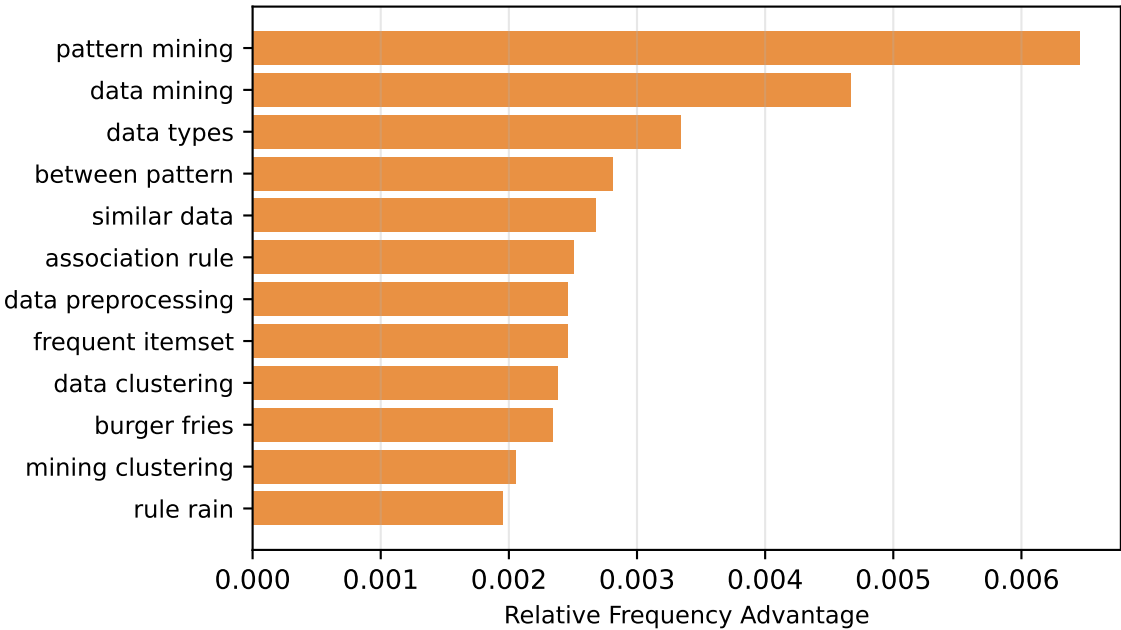
(B) Words More Common in LOW Root TTR
(Poor Responses)



(C) Bigrams More Common in HIGH Root TTR
(Rich Responses)

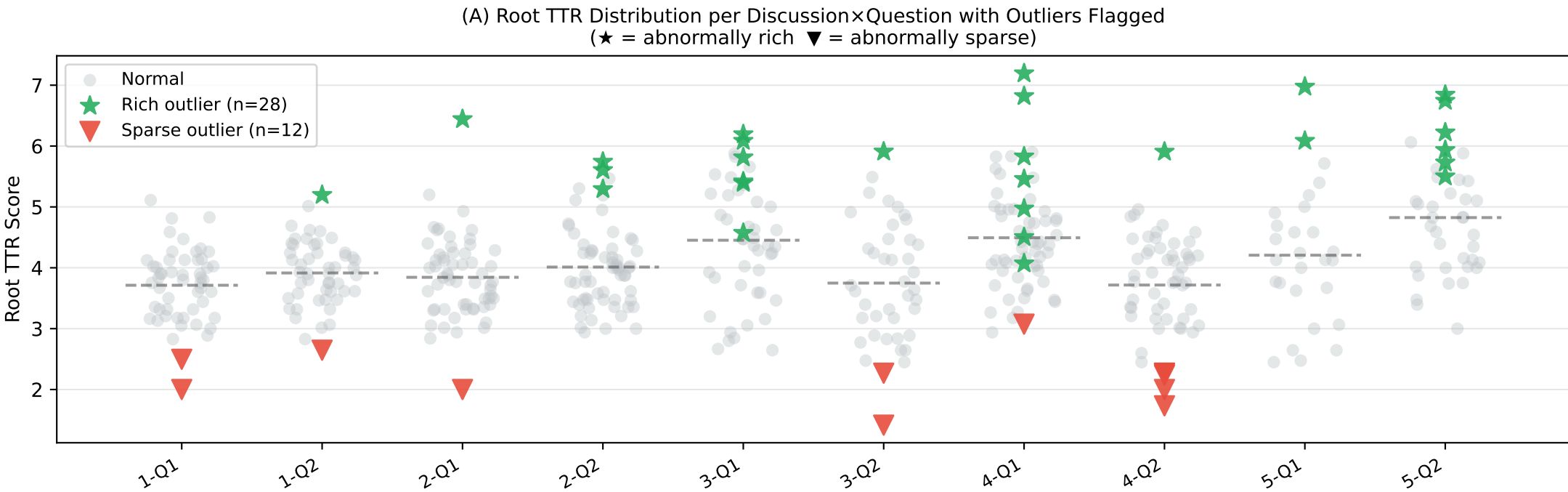


(D) Bigrams More Common in LOW Root TTR
(Poor Responses)



Key Finding: Rich responses use specific, varied terminology ("real", "real world"). Poor responses repeat generic words ("data", "pattern mining") — a pattern of conceptual narrowness, not just short length.

Outlier Detection: Responses with Abnormally High or Low Vocabulary Richness



(B) Most Vocabulary-Rich Outlier Responses

D-Q	Root TTR	Response Preview
4-Q1	7.194	Similarity: Both Pattern mining and clustering are methods
5-Q1	6.975	Outliers can be view as both positive and negative things b
5-Q2	6.841	When a stock's trading volume spikes 10 times in minutes,
4-Q1	6.820	Conceptual and difference between "pattern mining" and "c
5-Q2	6.742	I think whether an AI agent could identify outliers more eff.
2-Q1	6.441	Thought in is relatively important: deep representation lear

(C) Most Vocabulary-Sparse Outlier Responses

D-Q	Root TTR	Response Preview
3-Q2	1.414	Lift score
4-Q2	1.732	graph-based clustering
1-Q1	2.000	I wanted to place all classes in one day.
2-Q1	2.000	A data 100 visualization site was interesting
4-Q2	2.000	Hierarchical clustering. Like a tree of cluste
4-Q2	2.236	I think using graph. Keep its similarity is oth

Summary of Findings

"Which Discussion and Question Type Produces the Richest Student Vocabulary?"

Rank	Root TTR Rankings (Discussion × Question)	Mean Root TTR	Std	n
#1	5-Q2	4.825	0.922	39
#2	4-Q1	4.493	0.860	63
#3	3-Q1	4.453	0.980	47
#4	5-Q1	4.207	1.116	27
#5	2-Q2	4.012	0.681	62
#6	1-Q2	3.915	0.539	50
#7	2-Q1	3.844	0.676	59
#8	3-Q2	3.749	0.993	41
#9	4-Q2	3.717	0.835	55
#10	1-Q1	3.713	0.598	52

1. DATA UNDERSTANDING: Q2 consistently richer than Q1 across all discussions. Evaluation questions demand broader vocabulary than example-giving questions.

2. PATTERN MINING: Rich responses use specific compound terms ("real world"). Poor responses repeat generic words ("data"). Richness is about conceptual specificity, not just word count.

3. OUTLIER DETECTION: 40 multi-dimensional outliers detected. Rich outliers appear mostly in Q2 analysis questions. Sparse outliers tend to be short Q1 responses with minimal vocabulary.